



ÇANKAYA UNIVERSITY

CENG 408 – METHODOLOGY REPORT

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1.Methodology and System Approach

1.1 General Framework of the Methodological Approach

This study addresses pulmonary nodule analysis on thoracic computed tomography (CT) data through a staged and modular methodological framework. Rather than presenting a single-step and fully end-to-end system, the study is structured as a sequence of successive stages, including understanding the dataset, designing the preprocessing pipeline, obtaining nodule regions through segmentation, and constructing the classification approach based on these outputs. The main purpose of this approach is to ensure that each methodological decision is grounded not only in general assumptions, but also in concrete findings derived from the dataset and in outputs validated during intermediate stages.

Accordingly, the methodology of this study should be regarded as a staged framework rather than a single directly classification-oriented system. In the initial stage, a baseline classification pipeline was developed using XML-guided nodule localization, 2.5D slice representation, and a ResNet-18 classifier. In the later stage, the study evolved toward a segmentation-aware preprocessing structure based on CT–SEG matching, reconstructed masks, and ROI-centered sample generation. Thus, the methodological contribution of the study lies not only in baseline classification, but also in establishing a stronger data preparation foundation for subsequent classification and explainability modules.

1.2 Distinctive Aspects of the Study

One of the distinctive aspects of this study is that it treats the preprocessing stage not as an ordinary technical preparation step, but as one of the central methodological components of the research. In this study, the data preparation process was not limited to simply organizing the images before feeding them into the model; it was also considered an integral part of accurately identifying nodule regions, validating the segmentation outputs, and generating meaningful inputs for classification from these outputs. In this respect, the study differs from the conventional direct classification approach.

In the early stages of the study, the structure of the dataset was examined in detail, the relationships between CT images and annotations were analyzed, and an appropriate preprocessing pipeline was established to ensure the reliable extraction of nodule regions. During this process, DICOM metadata, CT–SEG matching, annotation geometry, and ROI selection were carefully evaluated. The findings showed that the dataset does not have a simple image-label structure; rather, it presents a multi-layered structure that requires careful alignment and validation. For this reason, the segmentation stage became one of the fundamental components of the study.

Another distinctive aspect of this study is that segmentation-oriented preprocessing and classification were treated as connected but methodologically separate stages. In the initial stage, classification experiments were conducted using XML-based localization and 2.5D inputs. In the later stage, segmentation-aware preprocessing was developed to obtain more reliable spatial representations of nodule regions through CT–SEG matching, reconstructed masks, and ROI extraction. In this way, the updated preprocessing pipeline was positioned not as a replacement for the earlier baseline classification stage, but as a stronger foundation for future classification and explainability studies.

Another distinctive feature of the study is its gradual development strategy. In the literature, there are many studies in which segmentation, classification, explainable artificial intelligence, and deployment components are presented simultaneously within a single system. However, in this study, the structure of the dataset was first understood, then segmentation-based nodule extraction was performed, and finally these outputs were used in the classification stage. In this way, each module was developed in a controlled manner and evaluated separately, preventing misleading results that might arise from the uncontrolled early integration of complex components.

1.3 Dataset Description: LIDC-IDRI

The primary dataset used in this study is LIDC-IDRI. LIDC-IDRI is one of the most widely used public datasets for pulmonary nodule analysis, malignancy classification, and artificial intelligence studies based on medical imaging. In its complete version, the dataset contains 1,018 patient cases; for each patient, thoracic CT data in DICOM format and XML-based annotations independently prepared by experienced radiologists are provided. This structure makes the dataset suitable for both classification and segmentation studies.

Within the scope of this project, the practical working environment focused particularly on the first 200 patients. In this subset, raw DICOM folders, metadata.csv, XML annotation files, and preprocess/cache outputs were evaluated together. The audit performed on the first 200 patients showed that there were 200 unique patients, 402 studies, and 3,796 metadata records. In more strict segmentation-oriented analyses, 192 unique CT series and 1,692 SEG annotation objects were identified. These results indicate that the dataset is not only usable, but also annotation-rich and provides a strong foundation for segmentation studies.

The reasons for selecting LIDC-IDRI include its multi-radiologist annotation structure, its status as an open-access and reproducible benchmark dataset, its compatibility with 2D, 2.5D, and 3D workflows, and its suitability for explainability-oriented studies. For these reasons, the dataset serves as a strong reference point not only for this study, but also for pulmonary nodule AI research in general.

1.4 Nodule Annotations and Label Definition

The XML annotations in the LIDC-IDRI dataset provide rich descriptive information for each nodule. These annotations include slice-based nodule contours, three-dimensional center coordinates (x,y,z), a malignancy score ranging from 1 to 5, and additional semantic features such as texture, margin, and sphericity. This information was particularly important for the

initial baseline classification stage of the study, where XML-derived nodule centers and malignancy scores were used to construct benign–malignant classification samples.

In the classification-oriented interpretation, only nodules measuring 3 mm or larger were included in the study. The reason for this is that nodules smaller than 3 mm are both visually and structurally more ambiguous, and therefore provide limited information for deep learning–based malignancy prediction. It has been reported that, in the complete version of the dataset, there are 2,669 annotated nodules measuring 3 mm or above. Each nodule may extend across multiple slices along the z-axis, and a single patient may contain more than one nodule.

During the malignancy labeling stage, the ordinal score ranging from 1 to 5 is converted into a binary classification label. In this conversion, scores of 1–2 are considered benign, while scores of 4–5 are considered malignant. Score 3, which represents uncertainty, is excluded from the dataset. In this way, label noise is reduced and a more reliable sample set is obtained for binary classification. Although this leads to a partially imbalanced data distribution, the issue is addressed during training through appropriate sampling strategies.

1.5 Structural Characteristics of the Dataset

One of the most important methodological findings of the study is that the dataset does not have a flat and simple structure. In conventional segmentation datasets, one would generally expect a single image volume and one corresponding mask. However, in this dataset, a patient may contain multiple studies, a study may include multiple series, a CT series may be associated with multiple SEG annotation objects, and related information in XML, SEG, and SR formats may exist simultaneously. For this reason, the dataset cannot be processed reliably by looking only at folder names or file order.

This directly affected the methodology. The initial baseline classification pipeline was established based on DICOM + XML logic, that is, by reconstructing the CT volume and selecting the nodule center from XML. However, more comprehensive audits showed that a stronger data preparation pipeline was necessary. As a result, the project evolved toward a segmentation-aware structure in which components such as DICOM UID references, SEG frame placement, and reconstructed mask generation became central.

Therefore, the complexity of the dataset was not treated as a difficulty, but rather as a fundamental reality that shaped the design of the system. The current approach develops a more reliable preprocessing, sample generation, and evaluation logic by taking this structural complexity into account.

1.6 Baseline Classification Pipeline

The first implemented learning pipeline in the project is a baseline classification framework developed for pulmonary nodules. At this stage, DICOM CT data and XML annotation files are used together. First, the DICOM slices are combined to reconstruct a three-dimensional CT volume, and voxel intensities are converted into Hounsfield Unit (HU) values. Then, a lung-focused windowing operation is applied; for example, the values are normalized using an

approximate HU range of -1000 to 400 and scaled to the $[0,1]$ range. After that, the nodule center coordinates are extracted from the XML files, and the central slice of the nodule is determined.

To preserve limited volumetric context without moving to full 3D processing, a 2.5D representation is used. In this approach, a fixed number of neighboring slices above and below the central slice are selected and stacked channel-wise to form a single input tensor. When necessary, an ROI crop centered on the nodule can also be applied to reduce background influence; for example, this crop may have dimensions such as 64×64 or 128×128 . Each nodule produces one training sample, and the final label is defined as benign (0) or malignant (1). After preprocessing and filtering, the study dataset is reported to consist of approximately 1,500–1,700 samples.

The classification model used in this baseline system is ResNet-18. Therefore, the baseline classification pipeline can be summarized as follows: DICOM + XML data usage, preprocessing and normalization, nodule-centered slice selection, 2.5D input generation, ResNet-18 classification, and performance evaluation using ROC-AUC and accuracy. At this stage, the system does not include segmentation or mask-based ROI extraction, but focuses directly on the classification problem. In addition, a user interface for displaying model outputs is also included in this stage. This baseline system represents the initial classification stage of the project rather than the final or current end-to-end methodology.

1.7 Rationale for Transitioning from a Classification-Oriented Approach to a Stronger Data Pipeline

Although the classification pipeline established at the beginning provided an understandable, practical, and reproducible foundation, more detailed data analyses showed that a representation based solely on classification was not sufficient to fully reflect the overall structure of the dataset. In particular, it was observed that the relationship between CT images and SEG annotations could be established reliably through DICOM metadata, and that full-volume mask generation was possible by placing SEG objects at the correct slice locations. This finding demonstrated that the dataset could be represented not only as center-based classification samples, but also through segmentation targets containing explicit spatial information.

This led to an important transformation in the methodology. As a result, beyond relying solely on XML centroid information to determine nodule location, the study explored stronger data preparation methods such as full-volume mask reconstruction, union bounding box calculation, and direct segmentation-based ROI extraction. This transition does not invalidate the previous classification stage; on the contrary, it strengthens the data preparation layer and makes it possible to develop classification, explainability, and other subsequent modules on a more solid foundation by benefiting from the outputs obtained through segmentation.

1.8 DICOM-Based CT–SEG Matching

One of the most critical technical steps of the updated methodology is establishing the correct relationship between CT series and SEG annotation objects. The audits showed that this matching cannot be performed reliably based on folder order, file names, or approximate assumptions. Instead, the matching is established directly through DICOM metadata. In particular, the SeriesInstanceUID field is used for CT, while the ReferencedSeriesSequence.SeriesInstanceUID field is used for SEG, and a CT–SEG pair list is created based on UID matching.

The analyses also showed that SEG objects are not stored directly as full-depth masks. While a CT volume may contain, for example, 133 or 261 slices, the corresponding SEG usually contains only a few frames in which the nodule appears. For this reason, in the methodology, SEG frames are placed at the appropriate CT slice indices to obtain a reconstructed full-volume mask. In this way, the mask volume reaches the same depth as the CT volume, making reliable ROI calculation possible at the slice level.

This DICOM-based matching forms the technical backbone of the new preprocessing pipeline. As a result, the annotation problem is transformed from being merely an approximate nodule localization task into a geometrically verifiable and reproducible representation problem.

1.9 ROI Analysis and Region-Centered Representation

Detailed ROI analyses showed that pulmonary nodules occupy a very small area within the original 512×512 CT plane. For this reason, training on full slices exposes the model to a large amount of unnecessary background and weakens the nodule signal. To reduce this problem, an ROI-based approach was adopted in the updated pipeline. In the segmentation-aware preprocessing stage, the ROI was defined by computing the union bounding box of the segmentation masks across the relevant slices, extending it with a fixed margin, and resizing the cropped region to a standard output size.

Different output sizes were evaluated for the ROI, including 128×128 , 160×160 , 192×192 , and 256×256 . As a result of these analyses, a margin of 24 pixels and an output size of 192×192 were determined to be the most balanced configuration. This setting preserves both the entire nodule and the surrounding anatomical context sufficiently, while also operating more efficiently than larger alternatives. In the earlier classification approach, ROI cropping was an optional and more approximate step; in the updated structure, however, ROI extraction has become a fundamental engineering decision validated by the data itself.

1.10 Input Representation: 2.5D Slice Stack

The common representation used in both the baseline classification pipeline and the updated segmentation-oriented pipeline is the 2.5D approach. This continuity is important because the 2.5D approach forms a methodological bridge between the initial and the current stages of the project. Full 3D processing was found to be computationally and memory intensive in the current experimental environment. On the other hand, single-slice 2D processing was

considered insufficient because it loses the contextual information that extends across neighboring slices. For this reason, the 2.5D approach was selected as a balanced solution between contextual information and computational cost.

In the updated representation, each sample consists of five slices: $[k-2, k-1, k, k+1, k+2]$. These slices are stacked channel-wise to form the input tensor. In the classification approach, the central slice is selected according to the XML centroid information, and the target is the benign–malignant label. In the segmentation-oriented approach, the central slice is selected from valid centers containing a positive mask, and the target becomes the mask of the central slice. Thus, while the 2.5D representation remains common in both methodological stages, only the supervision target changes.

1.11 Sampling Policy: Positive-Center-Only Approach

During the development of the segmentation-oriented preprocessing pipeline, different sample generation strategies were evaluated. Although generating independent samples from neighboring slices may initially seem like a way to increase the amount of data, experiments showed that this approach caused several problems. Among these issues were cases where the central target mask became empty, neighboring slices unnecessarily behaved as independent targets, and the dataset became filled with samples dominated by excessive background.

For this reason, the study adopted a positive-center-only sampling policy. According to this policy, a training sample is generated only when the central slice itself contains a positive mask. The neighboring slices are used only as contextual information and are not treated as independent targets. In this way, each sample has a meaningful central target, the sample structure remains more consistent, and unnecessary background dominance is reduced. The pilot preprocessing outputs also showed that this approach works correctly in practice.

1.12 Updated Preprocessing Pipeline

The updated preprocessing pipeline begins with the creation of a CT–SEG pair list using DICOM UID matching. Next, the CT volume is read, and its depth and z-range information are extracted. The corresponding SEG object is then read, and its frames are placed at the correct CT slice indices using DICOM references and spatial overlap information. In this way, a reconstructed mask volume with the same depth as the CT volume is obtained. In the next step, positive slices are identified, valid centers are selected according to the positive-center-only rule, and a five-slice 2.5D input is generated for each center. ROI cropping is applied using the union bounding box + 24-pixel margin rule, and the cropped images are resized to 192×192 . The resulting samples are stored as cache files; thus, the need to repeatedly read DICOM data during training is eliminated.

This pipeline was designed not only for accuracy but also for efficiency. Instead of resizing every slice in the entire CT volume, only the slices required to form valid 2.5D contexts are processed. This significantly reduces unnecessary computational load and I/O cost.

1.13 Pilot Preprocessing Validation

To verify whether the designed preprocessing logic was functioning properly, a pilot preprocessing study was conducted on 20 cases. In this pilot study, the CT depth, CT z-range, number of SEG frames, SEG z-range, mapped CT indices, reconstructed mask depth, positive slices, number of valid centers, number of generated 2.5D samples, number of resized slices, and processing time were reported for each case. These outputs demonstrated that the DICOM-based alignment worked correctly and that the SEG frames were placed at meaningful CT slice locations.

The pilot also provided a significant computational advantage. While the average CT depth was approximately 180 slices, only about 11 slices per case needed to be resized. This corresponds to an approximate 15.9-fold speedup in preprocessing. Thus, it was confirmed that the new preprocessing pipeline is not only technically correct, but also sufficiently efficient for practical training processes.

1.14 Cache Organization and Dataset Splitting

Since the dataset may contain multiple annotation objects for a single CT and multiple related samples for a single patient, storing the preprocessing outputs in an organized manner is an important design decision. In this study, the cache structure was built on an annotation-object basis. In other words, each cache file corresponds to one SEG annotation object. This structure both increases modularity and provides a data organization scheme compatible with the multi-annotator nature of the dataset.

When performing the training and validation split, a patient-based split was preferred. Slice-based or random sample-based splitting may cause highly related samples from the same patient to appear in both the training and validation sets, which would lead to information leakage. Patient-based splitting provides a more realistic evaluation of model performance.

1.15 Model Training Strategy

The model training strategy in this study reflects the staged methodological development of the project. In the initial stage, the objective was to establish a baseline benign–malignant classification pipeline using XML-based 2.5D inputs and a ResNet-18 classifier. At this stage, the main evaluation criteria were classification metrics such as ROC-AUC and accuracy. This baseline constituted the first functional and demonstrable classification framework of the project.

In the later stage, the focus shifted toward validating whether the updated preprocessing and data representation logic could produce learnable segmentation data. For this purpose, a SegResNet-oriented sanity-check training strategy using 2.5D representation was applied. This stage was not intended to present the final optimized model; rather, it was designed to verify that reconstructed masks, ROI cropping, positive-center-only sampling, and cache-based sample generation together formed a technically consistent and learnable data pipeline.

In the initial experiments, using 20 usable cache files and 146 samples, the tiny-overfit Dice score increased from approximately 0.284 to 0.793, while the best validation Dice in the mini-

training stage reached 0.713. In experiments conducted with larger datasets, this learnability was preserved, and the validation Dice values were reported to reach approximately 0.742 and 0.712.

Therefore, the baseline classification stage and the later segmentation-oriented validation stage should be interpreted as two connected but distinct methodological components. The former established the initial classification reference point of the study, whereas the latter provided a stronger data preparation foundation for subsequent classification, explainability, and model development studies.

1.16 Explainability

Explainable artificial intelligence (XAI) holds an important place in the conceptual design of the project. In particular, attention mechanisms and Grad-CAM-like methods are important in medical imaging because they can show which regions the model focuses on when making a decision. In the earlier methodology text, these components were presented not as implemented modules, but as conceptual components that could be added to the system in future stages. This approach is also maintained here. XAI is important in terms of clinical trust; however, at the current stage, it has not yet been implemented as an active module.

1.17 Summary of the Methodology

In summary, the methodology of this study develops along two interconnected main paths. The first path is the baseline classification pipeline, which consists of DICOM reconstruction, XML-based nodule localization, HU normalization, 2.5D slice stacking, ResNet-18 classification, and user interface-supported output presentation. The second path is the more recent data preparation pipeline, which is based on DICOM-based CT–SEG matching, reconstructed full-volume mask generation, ROI-centered cropping, five-slice 2.5D input, positive-center-only sampling, annotation-object-based cache structure, and patient-based splitting.

When these two paths are considered together, the methodology is not a structure composed of scattered and disconnected experiments; rather, it is a consistent system approach that has gradually strengthened through insights gained from the true nature of the dataset. While the initial classification stage provides a reproducible starting point, the newer segmentation-aware structure offers a technically stronger foundation for future model development.